

# Exact Spectral Leakage in Filtered Galois-Ring Channels

Frobenius-orbit accounting, hidden-state dependence,  
and robust adaptive control

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**Scope.** This paper studies a finite exact information-theory laboratory inspired by filtered Frobenius structures. It is not a construction of prismatic cohomology, a production cryptosystem, an asymptotic hardness claim, or a physical side-channel evaluation.

## Abstract

We develop an exact finite laboratory connecting a filtered Frobenius differential graded algebra, discrete Fourier transforms over Galois rings, Bayesian quantitative information flow, hidden-state inference, and adaptive channel design. Complete Frobenius orbits are treated as the atomic spectral components: conjugate frequencies are algebraic constraints rather than independent samples. Under independent component noise, Bayes vulnerability factorizes exactly; a latent state shared across components and queries destroys that factorization even when selected marginals are unchanged. We give a safe vector-valued Pareto elimination invariant, a strict executable counterexample to a two-endpoint shortcut outside its binary-complement hypothesis, and an exact Hamming-count transfer quotient for a three-state branching hidden Markov law. Translation reduction and complete-orbit sufficient states make exhaustive adaptive control possible without replacing integer likelihoods by floating-point approximations. Direct unreduced oracles agree with all optimized channels on normalization, true Bayes recovery, and two misspecified-decoder recoveries. Exhaustive finite policy censuses then separate fixed, scheduled, greedy, globally optimal, and minimax designs on cyclic groups of orders three, five, and seven. The contribution is the exact synthesis, its counterexamples, and its reproducible verification boundary; the constituent theories remain established prior work.

**Keywords:** quantitative information flow; Galois rings; finite Fourier transform; Frobenius orbit; hidden Markov model; Bayes vulnerability; adaptive leakage; exact inference.

## 1 Introduction

Spectral structure is simultaneously helpful and dangerous in information leakage models. A transform can expose low-dimensional coordinates that make a channel intelligible, but it can also tempt the analyst to count algebraically conjugate frequencies as independent evidence. Independence assumptions create a second trap: a channel that factorizes one query at a time can fail to factorize jointly when a hidden state is reused across components or time. Finally, an exact one-query decoder need not remain exact after an apparently minor change in the latent process.

This paper isolates those issues inside a deliberately small laboratory. The carrier is a finite filtered differential graded algebra over  $\mathbb{Z}/4\mathbb{Z}$ , with a characteristic-eight reuse control. Cyclic convolution is diagonalized over Galois rings. Binary masks and secrets are indexed by complete Frobenius orbits. Leakage is measured by posterior one-guess recovery, and temporal dependence ranges from fresh local noise to a full-support three-state hidden Markov law. Every state space is finite, so each claim can be checked by exact integer or rational arithmetic and, where tractable, against an independent unreduced oracle.

**Contributions.** Within the declared finite model, we:

1. connect a filtered Frobenius carrier to exact Galois-ring Fourier coordinates while treating complete Frobenius components as the mask and leakage units;
2. prove an exact product rule under component independence and retain executable witnesses showing why shared latent state invalidates it;
3. formulate a vector/Pareto elimination invariant that delays secret maximization until the complete latent mixture is assembled;
4. exhibit a strict channel-derived counterexample to a binary two-endpoint reduction once a third emission breaks its hypothesis;
5. derive a per-query leading/deep Hamming-count sufficient statistic for a branching three-state law and verify it against direct enumeration;
6. exhaustively evaluate finite adaptive policy families and nine-schedule robust searches, preserving exact ties, gaps, and decoder witnesses; and
7. release a sealed implementation with 126 regression tests and exact human- and machine-readable reports.

The paper is organized as follows. [Section 2](#) positions the work against its several parent literatures before any novelty claim is made. [Sections 3 to 5](#) construct the carrier, spectral geometry, and leakage metric. [Sections 6 and 7](#) develop the dependent channels and safe elimination rules. [Sections 8 and 9](#) give the finite policy experiments and exact results. Limitations, reproducibility, conclusions, derivations, and code follow.

## 2 Related Work and Novelty Positioning

The literature review was conducted before drafting the technical claims. It searched primary papers and publisher records through 4 September 2026 across algebraic geometry, Galois-ring Fourier analysis, coding theory, QIF, hidden Markov inference, adaptive experiment design, robust channel design, masking, and transform-domain side channels. [Table 1](#) summarizes the nearest neighbors and the boundary retained here.

Table 1: Nearest bodies of prior work and the paper’s narrower contribution.

| Area                  | Established result                                                                                      | Contribution retained here                                                                      |
|-----------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Prismatic theory      | Site-theoretic integral $p$ -adic cohomology with Frobenius and Nygaard structures                      | A finite prismatic-style organizing carrier only; no claim to compute that cohomology           |
| Finite Fourier theory | DFTs over rings, Galois-ring coordinates, cyclotomic cosets, and Frobenius descent                      | Complete components as exact leakage/mask units carried through dependent inference             |
| QIF                   | Bayes vulnerability, gain functions, axioms, composition, and dynamic secrets                           | Exact rational posteriors, fitted-decoder losses, and model-specific counterexamples            |
| Graphical inference   | HMM forward recursion, sum-product, and variable elimination                                            | A specialized positive-vector frontier and count-transfer quotient with direct equality oracles |
| Adaptive design       | Sequential experiments, adaptive side channels, Bellman recursions, leakage games, channel optimization | Exact census of declared algebraic action trees and finite minimax schedules                    |
| Side channels         | Power/EM/timing models, probing security, masking, and NTT attacks                                      | Motivation only; no physical-trace or implementation-security claim                             |

## 2.1 Prismatic structures

Prismatic cohomology is an integral  $p$ -adic cohomology theory built from prisms and the prismatic site, with Frobenius and Nygaard structures [1]. Its relationship with topological Hochschild and cyclic homology belongs to a much richer homotopical setting [2, 3]. Our four-generator finite algebra borrows only the motifs “filtration plus Frobenius.” Calling it prismatic cohomology would be mathematically false; throughout, *prismatic-style* names an organizational analogy and nothing more.

## 2.2 Finite-ring Fourier analysis and orbit descent

Finite rings supporting a DFT have long-standing structural theory [4]. Galois rings are central in quaternary coding [5], and explicit bases and Fourier transforms over them are well studied [6, 7]. Most directly, Kumallagov et al. [8] prove finite-field Galois descent of Fourier spectra and the optimality of orbit-seed coordinates. We therefore make no priority claim for Frobenius consistency or cyclotomic-orbit compression. The new role assigned here is informational: an entire orbit, rather than each frequency, is the primitive unit passed through mask design, posterior recovery, hidden-state coupling, and adaptive control.

## 2.3 Quantitative information flow

Conditional min-entropy and Bayes vulnerability were established as QIF measures by Smith [9]. Generalized gain functions and axiomatic treatments followed [10, 11]; product channels and compositional bounds are treated by Kawamoto et al. [12]. Dynamic secrets and abstract hidden Markov models already connect temporal state and information flow [13, 14]. A recent Bayesian leakage framework handles inference attacks and auxiliary information in cryptographic leakage models [15]. We use the standard Bayes quantity and claim novelty only for exact evaluation on the present filtered spectral channel, including its correct-versus-fitted decoder comparisons.

## 2.4 Graphical inference, adaptive attacks, and robust design

Finite-state forward inference is classical [16], as are sum-product and exact variable elimination [17, 18]. Sequential design and active hypothesis testing predate this work by decades [19, 20]. Adaptive side-channel strategies and Bellman recursions for generic leakage functions are also established [21, 22], as are optimal and deterministic minimum-leakage channel design [23, 24]. Our backward induction is not a new dynamic-programming principle. The contribution is an exact specialized state reduction whose full finite tree family can be enumerated with retained witnesses.

## 2.5 Side-channel boundary and scoped novelty statement

Physical side-channel analysis includes differential power analysis, probabilistic attack evaluation, probing models, masking, and direct attacks on the number theoretic transform [25–28]. The present observations are algebraic random variables, not measured traces. Consequently the results do not imply implementation resistance or an attack on a deployed system.

The search found no primary source that combined a filtered characteristic- four/eight carrier, Galois-ring DFTs with complete-orbit mask accounting, Bayes leakage under cross-component and cross-query hidden state, executable shortcut counterexamples, fitted-decoder witnesses, and exhaustive finite adaptive/minimax censuses. We are therefore not aware of a previous treatment of that complete combination. This is a scoped literature-search finding, not an absolute priority guarantee.

### 3 Finite Filtered Frobenius Carrier

Let  $A = \mathbb{Z}/4\mathbb{Z}$  and define the graded-commutative algebra

$$C = A[x, \eta]/(x^2, \eta^2), \quad |x| = 0, \quad |\eta| = 1, \quad (1)$$

with differential and Frobenius-like endomorphism

$$d(x) = 2\eta, \quad d(\eta) = 0, \quad (2)$$

$$\varphi(a + bx + c\eta + d'x\eta) = a + 2bx + 2c\eta. \quad (3)$$

Modulo two,  $\varphi$  is squaring, and direct evaluation on the four basis monomials verifies  $d\varphi = \varphi d$ . Let  $J = (2, x, \eta)$  and  $F^r C = J^r$ . The complete filtration is

$$|F^0 C| = 256, \quad |F^1 C| = 128, \quad |F^2 C| = 16, \quad |F^3 C| = 2, \quad |F^4 C| = 1, \quad (4)$$

with

$$F^1 = (2, x, \eta), \quad F^2 = (2x, 2\eta, x\eta), \quad F^3 = (2x\eta).$$

The differential raises filtration, and a direct cycle/boundary calculation gives

$$H^0(C) \cong \mathbb{Z}/4\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}, \quad H^1(C) \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}. \quad (5)$$

Two filtration classes form the binary leakage carrier:

$$h = 2x \in F^2 \setminus F^3, \quad t = 2x\eta \in F^3. \quad (6)$$

Each cyclic coordinate therefore has a leading bit  $h$  and a deeper bit  $t$ . A characteristic-eight control replaces  $A$  by  $\mathbb{Z}/8\mathbb{Z}$ , takes  $d(x) = 4\eta$ , and uses  $h = 4x$ ,  $t = 4x\eta$ ; it preserves the two-layer binary carrier while increasing the filtration depth.

*Remark 3.1* (Interpretive boundary). Equations (1)–(6) are an explicit finite DGA and its ideal-adic filtration. They are not a prism, a delta-ring construction, or a computation of prismatic cohomology.

### 4 Galois-Ring Fourier Geometry

Let  $m$  be odd and choose a Galois ring  $E = \text{GR}(4, d)$  containing a Teichmüller unit  $\zeta$  of order  $m$ . For  $f = (f_0, \dots, f_{m-1}) \in A^m$ , define

$$\widehat{f}(k) = \sum_{n=0}^{m-1} f_n \zeta^{-kn}, \quad f_n = m^{-1} \sum_{k=0}^{m-1} \widehat{f}(k) \zeta^{kn}. \quad (7)$$

The inverse is exact because odd  $m$  is a unit in  $A$ . If  $\sigma$  is the Galois-ring Frobenius, base-ring inputs satisfy

$$\sigma(\widehat{f}(k)) = \widehat{f}(2k \bmod m). \quad (8)$$

Thus a complete doubling orbit  $\mathcal{O}(k) = \{k, 2k, 4k, \dots\}$  is one constrained spectral component, not  $|\mathcal{O}(k)|$  unrelated observations. For the three groups used below,

$$\begin{aligned} C_3 &: (0), (1, 2), \\ C_5 &: (0), (1, 2, 4, 3), \\ C_7 &: (0), (1, 2, 4), (3, 6, 5). \end{aligned} \quad (9)$$

**Proposition 4.1** (Complete-orbit rank). *For a binary residue vector on  $C_m$ , the view consisting of every coefficient in a doubling orbit  $\mathcal{O}$  has  $\mathbb{F}_2$ -rank  $|\mathcal{O}|$ . All other coefficients in that component are Frobenius conjugates of one orbit seed.*

*Proof.* Equation (8) makes the conjugacy constraint explicit. Choosing a normal or polynomial basis for the residue field gives  $|\mathcal{O}|$  base-field coordinates for the seed. The implementation independently enumerates every binary input and row-reduces each complete view; the measured ranks are 1, 2 for  $C_3$ , 1, 4 for  $C_5$ , and 1, 3, 3 for  $C_7$ .  $\square$

For a union  $M$  of complete orbits, let  $U_M$  be the binary mask subspace supported on those components. Then

$$\dim_{\mathbb{F}_2} U_M = \sum_{\mathcal{O} \subseteq M} |\mathcal{O}|. \quad (10)$$

In the noiseless quotient channel with a uniform  $m$ -bit secret, the posterior is a coset of size  $2^{\dim U_M}$ . Hence its one-guess recovery is  $2^{-\dim U_M}$ , and the revealed binary dimension is  $m - \dim U_M$ . A direct 16-design census across the declared small groups retains three orbit-complete passing designs and rejects every partial-orbit mask.

## 5 Leakage Channel and Exact Metrics

Let  $S$  be a finite secret with prior  $\pi_S$ , and let  $K(y | s)$  be a finite channel. The posterior Bayes vulnerability is

$$V_B(S | Y) = \sum_y \max_s \pi_S(s) K(y | s). \quad (11)$$

This is the probability of optimal one-guess recovery. Conditional min-entropy is  $H_\infty(S | Y) = -\log_2 V_B(S | Y)$ ; leakage is the prior-to-posterior decrease. The laboratory also computes total variation, collision quantities, and full posterior supports, but (11) is the optimization objective.

Each complete orbit of dimension  $d$  carries two  $d$ -bit secret layers, so its local module has  $2^{2d}$  states. Writing a local secret as  $(s_0, s_1)$ , the three public bijections are

$$A_{\text{std}}(s_0, s_1) = (s_0, \text{shift}_2(s_0) + s_1), \quad (12)$$

$$A_{\text{diag}}(s_0, s_1) = (s_0, s_1), \quad (13)$$

$$A_{\text{shr}}(s_0, s_1) = (s_0, s_0 + s_1), \quad (14)$$

where addition is binary and  $\text{shift}_2$  is cyclic shift by two positions. Query  $i$  returns

$$Y_i = A_i S + E_i. \quad (15)$$

If the first action is invertible, later histories admit the exact translation

$$D_i = Y_i + A_i A_1^{-1} Y_1 = E_i + A_i A_1^{-1} E_1, \quad i \geq 2. \quad (16)$$

The secret translation removes  $S$  from the canonical difference state; each reduced state represents  $|S|$  raw histories. This is a sufficient state reduction, not an approximation.

**Theorem 5.1** (Independent component product rule). *Suppose  $S = (S_j)_j$  has a product prior and, conditionally on  $S$ , the complete-orbit channels are independent:  $K(y | s) = \prod_j K_j(y_j | s_j)$ . Then*

$$V_B(S | Y) = \prod_j V_B(S_j | Y_j). \quad (17)$$

*Proof.* For fixed  $y$ , the maximization separates over the Cartesian product of local secrets. Summation over the Cartesian product of outputs then factorizes. This is the equality specialization of familiar product-channel composition; the stated hypotheses are essential.  $\square$

## 6 Shared Hidden State Across Components and Queries

A globally shared latent process replaces the product channel by

$$K(y | s) = \sum_{z_{1:q}} p(z_{1:q}) \prod_{i=1}^q \prod_j K_{ij}(y_{ij} | s_j, z_i), \quad (18)$$

which generally cannot be inserted into (17). The project progressively tests a fresh component-local binary state, persistent orbit-local and globally shared binary states, a deterministic ternary state, and the following branching ternary law.

Let

$$\pi_Z = \frac{1}{6}(3, 2, 1), \quad T = \frac{1}{4} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}. \quad (19)$$

State zero applies no common flip; state one complements leading and deep layers; state two complements only the leading layer. The state is shared by all complete components at query  $i$ . Conditional on the entire path, base bits are independent. The exact marginals through query four are

$$\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right), \left(\frac{3}{8}, \frac{1}{3}, \frac{7}{24}\right), \left(\frac{11}{32}, \frac{1}{3}, \frac{31}{96}\right), \left(\frac{43}{128}, \frac{1}{3}, \frac{127}{384}\right). \quad (20)$$

For a complete secret candidate, put  $D_i(s) = \text{diag}(E_i(0;s), E_i(1;s), E_i(2;s))$ . The exact likelihood is

$$L(s) = \pi_Z^T D_1(s) T D_2(s) \cdots T D_q(s) \mathbf{1}. \quad (21)$$

It is evaluated in two independent ways: explicit enumeration of all  $3^q$  paths, and the forward recursion

$$\begin{aligned} \alpha_1(z) &= \pi_Z(z) E_1(z), \\ \alpha_i(z) &= E_i(z) \sum_y \alpha_{i-1}(y) T(y, z), \quad L(s) = \sum_z \alpha_q(z). \end{aligned} \quad (22)$$

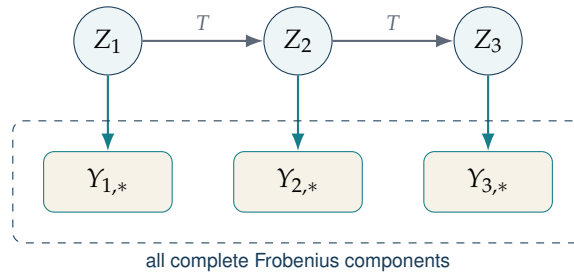


Figure 1: One latent state is shared by every complete spectral component at a query. Conditioning on the full path restores base-bit independence.

The pairwise cross-query controls are  $\text{Cov}(h_1, h_2) = 1/16$ ,  $\text{Cov}(t_1, t_2) = 1/18$ , and  $\text{Cov}(h_1, t_2) = 1/24$ . These moments make the temporal memory visible, but the decoder always uses the full likelihood (21).

## 7 Exact Elimination Invariants

### 7.1 Positive-vector Pareto elimination

At an intermediate separator, associate each local candidate with a nonnegative likelihood vector  $v = (v_1, \dots, v_r)$ , indexed by the latent states or paths not yet marginalized. Define  $v \preceq w$  when  $v_k \leq w_k$  for every  $k$ .



**Proposition 7.1** (Safe dominance pruning). *If every later coefficient applied to a separator vector is nonnegative, a candidate  $v$  with  $v \preceq w$  can be removed: no positive continuation and latent prior can make  $v$  strictly outperform  $w$ .*

*Proof.* Every remaining operation before the secret maximum is a coordinatewise product by a nonnegative emission, multiplication by a nonnegative transition matrix, or addition. These operations are monotone in every coordinate. Therefore dominance is preserved to the final positive scalar score.  $\square$

The rule does *not* permit independently maximizing the coordinates and then mixing them; those maxima can belong to incompatible secrets. The implementation carries the exact Pareto frontier until all complete components for the relevant latent separator have been assembled.

## 7.2 Strict failure of a two-endpoint shortcut

A binary complement model possesses a special constant-product relation, under which the final score can be determined by two extrema. That shortcut is not a general property of hidden-state channels. In the actual  $C_3$  channel, a three-state emission yields the Pareto vectors

$$\begin{aligned} v^{(1)} &= (9, 117649, 1), \\ v^{(2)} &= (1058841, 1, 117649), \\ v^{(*)} &= (3969, 1750329, 35721). \end{aligned}$$

The old shortcut retains  $v^{(1)}$  and  $v^{(2)}$  as the minimum and maximum of its designated coordinate. The interior vector  $v^{(*)}$  dominates  $v^{(1)}$  and is incomparable with  $v^{(2)}$ , so it belongs to the true Pareto frontier. With cleared prior weights  $(3, 2, 1)$ ,

$$\max_{j \in \{1,2\}} (3, 2, 1) \cdot v^{(j)} = 3,294,174 < 3,548,286 = (3, 2, 1) \cdot v^{(*)}. \quad (23)$$

Thus retaining only the old endpoints discards the true optimizer. The witness is executable and arises from the declared channel, rather than from an unrelated synthetic vector set.

## 7.3 Count-transfer sufficient statistic

Let query  $i$  have total leading and deep Hamming weights  $(a_i, b_i)$  across all complete components, with  $n = m$  binary coordinates per layer. Base bit weights are  $(3, 1)$  for the leading layer and  $(7, 1)$  for the deep layer. After clearing denominators, the state-conditioned emissions are

$$\begin{aligned} E_i(0) &= 3^{n-a_i} 7^{n-b_i}, \\ E_i(1) &= 3^{a_i} 7^{b_i}, \\ E_i(2) &= 3^{a_i} 7^{n-b_i}. \end{aligned} \quad (24)$$

**Theorem 7.2** (Exact count-transfer quotient). *For the emission family (24), the sequence  $((a_i, b_i))_{i=1}^q$  is sufficient for the complete hidden-state likelihood of a secret candidate. Component count codes compose by integer addition, and the resulting transfer score equals explicit latent-path enumeration.*

*Proof.* Conditional base-bit weights are products. Within each query and layer, a coordinate contributes only according to whether it agrees with the unflipped or complemented base error. The products therefore depend on coordinates only through  $a_i$  and  $b_i$ , giving (24). Substitution in (21) proves sufficiency. Expanding the matrix product recovers the sum over all  $3^q$  paths term by term.  $\square$



The quotient is specific to the declared exchangeable Bernoulli-complement emissions. It is not asserted for arbitrary HMM observations. A second invalidation witness also matters: under the full-support transition,

$$\begin{aligned} \text{standard} &\rightarrow \text{diagonal} \rightarrow \text{unit-shear}, \\ \text{standard} &\rightarrow \text{unit-shear} \rightarrow \text{diagonal} \end{aligned}$$

have the same tail-action multiset but distinct Bayes numerators, 1,284,369,637,867,812 and 1,273,560,535,783,140. Time ordering cannot be cached away.

## 8 Adaptive and Robust Finite Control

The action set is the three maps in (14), with the first query fixed to  $A_{\text{std}}$ . A fixed policy repeats standard. A scheduled policy follows a declared open-loop tail. A greedy policy optimizes the next stage only, whereas a globally optimal policy is obtained by backward induction. The two algorithms and all tie sets are computed separately even when their final values coincide.

For  $C_5$  and  $C_7$  at depth three, the complete four-state DC prefix selects the third action. The family therefore contains

$$3^{1+4} = 243 \tag{25}$$

implicit trees and requires 36 exact continuations. For the complete  $C_3$  carrier at depth four, four depth-two DC states and sixteen depth-three DC states select later actions, giving

$$3^{1+4+16} = 3^{21} \tag{26}$$

implicit trees. Translation reduction stores 21 decisions and evaluates 432 last-step continuations over  $64^3$  sufficient histories, rather than enumerating  $64^4$  raw histories for every policy.

The robust experiment fixes the first action and enumerates all nine ordered second/third action pairs. Each is scored under five true laws:

1. fresh component-local binary state;
2. persistent orbit-local binary state;
3. persistent globally shared binary state;
4. deterministic ternary Markov state; and
5. branching ternary hidden state (19).

For every schedule, the report retains every worst-law tie, exact tier and gap, a worst-transcript witness, and disagreement witnesses for persistent-binary and deterministic-ternary fitted decoders evaluated under the true branching law. “Robust” means minimax only over this declared finite family.

## 9 Exact Results

### 9.1 Direct equality oracles

The unreduced channels enumerate every translated transcript and complete secret without calling the count-transfer engine. Table 2 reports the shared exact results. “Binary fitted” maximizes a persistent globally shared binary model; “deterministic fitted” maximizes the preceding deterministic ternary model. Both choices are evaluated under the true branching channel. All direct and reduced routes normalize to one.

The strict mismatch gaps demonstrate a point stronger than simple parameter error: a decoder can use the correct observation alphabet and plausible marginals, yet choose a different secret because it fits the wrong temporal or cross-component dependence. Each

Table 2: Direct enumeration equals count transfer exactly. Parentheses give decimal values only for readability; the computation uses the fractions.

| Group | $q$ | Direct   | True branching Bayes                                  | Binary fitted | Deterministic fitted |
|-------|-----|----------|-------------------------------------------------------|---------------|----------------------|
| $C_3$ | 3   | $64^3$   | $107030803155651 \times 2^{-48}$<br>(0.380249798)     | 0.210936734   | 0.188159799          |
| $C_3$ | 4   | $64^4$   | $8363572086883399839 \times 2^{-64}$<br>(0.453390151) | 0.200301680   | 0.175176733          |
| $C_5$ | 2   | $1024^2$ | $51129052896991 \times 2^{-48}$<br>(0.181646886)      | 0.130019691   | 0.110470326          |

public policy stores at least one transcript at which that disagreement occurs, together with true posteriors and exact filtering/smoothing distributions.

## 9.2 Adaptive policies

Table 3 gives the branching-law recovery values. For  $C_5$  and  $C_7$ , scheduled, greedy, and global policies uniquely choose diagonal as the second action and tie in final value. At  $C_3, q = 4$ , greedy and global also tie, but strictly exceed the scheduled policy. Equality of final values is not used to identify their algorithms or decision trees.

Table 3: Exact branching-law Bayes recovery for representative policy families.

| Control      | Policy                      | Exact recovery                                     | Decimal     |
|--------------|-----------------------------|----------------------------------------------------|-------------|
| $C_5, q = 3$ | fixed                       | $12479777669624554355511 \times 2^{-76}$           | 0.165168482 |
|              | scheduled / greedy / global | $6349567569731380852239 \times 2^{-74}$            | 0.336143308 |
| $C_7, q = 3$ | fixed                       | $37257119202207049319536034625683 \times 2^{-108}$ | 0.114807362 |
|              | scheduled / greedy / global | $98702132900086285618377508839379 \times 2^{-108}$ | 0.304149429 |
| $C_3, q = 4$ | fixed                       | $2711541269830791513 \times 2^{-63}$               | 0.293985894 |
|              | scheduled                   | $8363572086883399839 \times 2^{-64}$               | 0.453390151 |
|              | greedy / global             | $8459390934058791939 \times 2^{-64}$               | 0.458584501 |

The depth-four engine records 1,814,640 component-count evaluations and 1,366,069,248 exact ordinal-rank lookups. Its retained Pareto frontiers total 65,792 candidates before pruning, 63,744 afterward, with maximum size 16. The large rank-lookup count is array indexing over exact integer orderings; all likelihood scores and posterior quantities remain arbitrary-precision integers or Fraction values.

## 9.3 Five-law robust schedule census

Both  $C_5$  and  $C_7$  produce four exact risk tiers with schedule ranks  $(1, 2, 2, 2, 3, 3, 3, 4, 4)$ . Exactly one of nine schedules is accepted:

$$\text{standard} \rightarrow \text{standard} \rightarrow \text{standard}.$$

The persistent globally shared binary law is uniquely worst for every schedule. The accepted worst-case recoveries are

$$\begin{aligned} C_5 : \quad & \frac{11874641829239003125}{36893488147419103232} \approx 0.321862812, \\ C_7 : \quad & \frac{519613248956875545739140625}{2475880078570760549798248448} \approx 0.209870120. \end{aligned} \quad (27)$$

This outcome should not be read as a universal recommendation to repeat the standard action. It is the minimax answer for the stated groups, priors, emissions, action alphabet, horizon, and five-law uncertainty set.

## 10 Reproducibility, Validation, and Limitations

The reference artifact is Prismatic–Fourier Lab v0.10.0. Its annotated tag dereferences to commit `f1035c36d572e7d245c54d38d6fe2c8367493ba7`; the release archive’s SHA-256 digest is `56dcafc72c298fbbd5035e2ce8f3891a40c50a5a7ca56c7055187248ef0098c3`. The public CLI emits both a human report and a 782,964-byte JSON record.

Validation has three layers. First, algebra, filtration, Galois arithmetic, DFT inversion, convolution, and orbit ranks are exhaustively checked on the finite carriers. Second, direct channels independently reproduce the reduced Bayes values, fitted recoveries, and normalizers in Table 2. Third, 126 regression tests freeze every milestone boundary. The author’s Windows runs reported 12/12 dedicated Milestone-10 tests and 126/126 complete tests; independent clean-extraction runs reproduced 126/126. Long runtimes are expected because the tests deliberately recompute exact large-group oracles.

The principal limitations are structural:

- groups, rings, actions, horizons, and uncertainty sets are finite and declared rather than asymptotic;
- the DGA is an analogy-driven finite carrier, not geometric prismatic cohomology;
- noise is algebraic and exchangeable inside layers, not calibrated to a device;
- the count quotient depends on the Bernoulli-complement emission family;
- exact enumeration establishes the stated instances, not computational hardness; and
- the literature search supports a defensible novelty position but cannot certify universal priority.

## 11 Conclusion and Further Work

The laboratory shows why spectral leakage analysis needs three kinds of bookkeeping at once. Algebraic bookkeeping groups conjugate Fourier coefficients into complete Frobenius components. Probabilistic bookkeeping keeps shared latent state intact until all components and queries have been combined. Decision-theoretic bookkeeping distinguishes the true Bayes decoder, fitted decoders, greedy choices, global backward induction, and finite minimax design. Dropping any one of these distinctions produces concrete errors in the declared examples.

The exact reductions are as important as the counterexamples. Secret translation replaces raw histories by canonical differences; component independence yields a genuine product theorem when its hypotheses hold; positive-vector dominance safely prunes hidden-state likelihoods; and the count-transfer theorem replaces coordinate configurations by per-query Hamming totals for the specified emission family. Direct oracles establish that these are equality-preserving transformations, not numerical heuristics.

Several extensions are natural. The algebraic side should move from cyclic groups to general finite abelian groups and from the selected Galois rings to broader finite local rings. The probabilistic side should characterize when a multi-state emission admits endpoint compression and when its count statistic is minimal sufficient. Symbolic decision diagrams, semiring compilation, or proof-producing elimination could extend the exact horizon while preserving witnesses. A proof-assistant formalization could connect the filtration, Fourier, posterior, and policy invariants in one checked development. Finally, the most consequential empirical step is to compare the model’s orbit-level rankings with measured FFT/NTT traces. Such a study would test whether this finite laboratory predicts anything about physical leakage; until then, it should remain what it is: a transparent, falsifiable information-theory model.

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## A Algebraic and Fourier Derivations

### A.1 Cohomology of the finite DGA

Write a degree-zero element as  $a + bx$ . By (2),  $d(a + bx) = 2b\eta$ , so it is a cycle exactly when  $b \in \{0, 2\}$ . There are no degree-minus-one boundaries, giving  $H^0 \cong \mathbb{Z}/4\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ . A degree-one element is  $c\eta + d'x\eta$ ; it is always a cycle because  $\eta^2 = 0$ . Boundaries are the two-element subgroup generated by  $2\eta$ , so  $H^1 \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$ . Both groups have eight elements.

The ideal powers follow from  $2^2 = x^2 = \eta^2 = 0$ :

$$\begin{aligned} J &= (2, x, \eta), \\ J^2 &= (2x, 2\eta, x\eta), \\ J^3 &= (2x\eta), \\ J^4 &= 0. \end{aligned}$$

Counting coefficient choices yields (4).

### A.2 DFT inversion and Frobenius consistency

Orthogonality of a primitive odd-order Teichmüller root gives

$$\sum_{k=0}^{m-1} \zeta^{k(n-r)} = \begin{cases} m, & n = r, \\ 0, & n \neq r. \end{cases}$$

Since  $m$  is a unit modulo four, substitution proves the inverse in (7). For coefficients fixed by  $\sigma$ ,

$$\sigma(\hat{f}(k)) = \sum_n f_n \sigma(\zeta^{-kn}) = \sum_n f_n \zeta^{-2kn} = \hat{f}(2k),$$

which proves (8).

### A.3 Why a shared latent state blocks the product rule

Under independence, maximization separates:

$$\max_{(s_j)_j} \prod_j q_j(s_j, y_j) = \prod_j \max_{s_j} q_j(s_j, y_j).$$

With a global latent variable, the relevant expression is instead

$$\max_{(s_j)_j} \sum_z p(z) \prod_j q_j(s_j, y_j, z).$$

The same  $z$  couples every factor inside the sum, so no algebraic identity permits replacing this by a product of local latent mixtures. Component marginals do not determine the joint vulnerability.

## B Exact Result Tables

## C Algorithms

### C.1 Exact complete-channel evaluation

For every translated transcript, the reference implementation conceptually performs the following operations:



Table 4: Complete Frobenius components and local module sizes.

| Group | Orbits                      | Residue dimensions | Two-layer module sizes |
|-------|-----------------------------|--------------------|------------------------|
| $C_3$ | $(0), (1, 2)$               | 1, 2               | 4, 16                  |
| $C_5$ | $(0), (1, 2, 4, 3)$         | 1, 4               | 4, 256                 |
| $C_7$ | $(0), (1, 2, 4), (3, 6, 5)$ | 1, 3, 3            | 4, 64, 64              |

Table 5: Finite adaptive families in the final branching experiment.

| Control    | Horizon | Implicit trees | Continuations | Branch state                                     |
|------------|---------|----------------|---------------|--------------------------------------------------|
| $C_5, C_7$ | 3       | $3^5 = 243$    | 36            | Four complete DC states                          |
| $C_3$      | 4       | $3^{21}$       | 432           | Four depth-two and sixteen depth-three DC states |

1. enumerate local errors separately inside every complete Frobenius component;
2. convert each local candidate to per-query leading/deep Hamming counts;
3. add count codes across all complete components;
4. evaluate the resulting type under the selected exact hidden-state law;
5. maximize only after the complete secret score exists; and
6. accumulate the chosen true-channel score into (11).

For a misspecified decoder, step 4 ranks candidates with the fitted law, while step 6 evaluates its selected secret under the true law.

## C.2 Backward induction

At the last decision layer, each sufficient prefix stores the action attaining the maximum exact continuation value and the full tie set. These values are summed into their parent prefixes, and the procedure repeats to query two. The greedy routine instead maximizes the immediate next-query value. Separate code paths and assertions prevent an accidental alias between the two policies.

## D Minimal Exact Python Kernel

The listing below is a compact, dependency-free transcription of the two central numerical invariants: latent-path enumeration equals transfer-matrix recursion, and only coordinatewise-dominated likelihood vectors are removed. The sealed project contains the complete orbit, policy, report, and witness machinery.

```

1  """Minimal exact kernel used by the paper's Appendix D.
2
3  This is a compact, dependency-free transcription of the likelihood and
4  dominance invariants in Prismatic--Fourier Lab v0.10.0. It is illustrative,
5  not a replacement for the sealed experiment package.
6  """
7
8  from __future__ import annotations
9
10 from fractions import Fraction
11 from itertools import product
12 from typing import Iterable, Sequence
13
14 Vector = tuple[int, ...]
15 Counts = tuple[tuple[int, int], ...]
```



```

16
17 PI = (3, 2, 1)
18 T = (
19     (2, 1, 1),
20     (1, 2, 1),
21     (1, 1, 2),
22 )
23
24
25 def emission_vector(n: int, leading: int, deep: int) -> Vector:
26     """Return cleared state-conditioned weights for one query.
27
28     ‘‘leading’’ and ‘‘deep’’ are Hamming totals across every complete
29 Frobenius component. The leading and deep base weights are (3,1) and
30 (7,1), respectively.
31     """
32
33     if not (0 <= leading <= n and 0 <= deep <= n):
34         raise ValueError("Hamming counts must lie in [0,n]")
35     return (
36         3 ** (n - leading) * 7 ** (n - deep),
37         3**leading * 7**deep,
38         3**leading * 7 ** (n - deep),
39     )
40
41
42 def transfer_score(n: int, counts: Counts) -> int:
43     """Evaluate  $\pi^T D_{-1} T D_{-2} \dots T D_{-q} 1$  with integer weights."""
44
45     if not counts:
46         raise ValueError("At least one query is required")
47     alpha = tuple(p * e for p, e in zip(PI, emission_vector(n, *counts[0])))
48     for pair in counts[1:]:
49         emission = emission_vector(n, *pair)
50         alpha = tuple(
51             emission[z] * sum(alpha[y] * T[y][z] for y in range(3))
52             for z in range(3)
53         )
54     return sum(alpha)
55
56
57 def path_score(n: int, counts: Counts) -> int:
58     """Independent  $3^q$  latent-path enumeration oracle."""
59
60     emissions = tuple(emission_vector(n, *pair) for pair in counts)
61     total = 0
62     for path in product(range(3), repeat=len(counts)):
63         weight = PI[path[0]] * emissions[0][path[0]]
64         for i in range(1, len(path)):
65             weight *= T[path[i - 1]][path[i]] * emissions[i][path[i]]
66         total += weight
67     return total
68
69
70 def pareto_frontier(vectors: Iterable[Vector]) -> tuple[Vector, ...]:
71     """Remove only vectors dominated in every latent coordinate."""
72
73     unique = sorted(set(vectors))
74     if unique and any(len(v) != len(unique[0]) for v in unique):
75         raise ValueError("All vectors must have equal dimension")
76     kept = []
77     for candidate in unique:

```

```

78         dominated = any(
79             candidate != other
80             and all(x <= y for x, y in zip(candidate, other))
81             for other in unique
82         )
83         if not dominated:
84             kept.append(candidate)
85     return tuple(kept)
86
87
88 def bayes_vulnerability(
89     prior: Sequence[Fraction],
90     channel: Sequence[Sequence[Fraction]],
91 ) -> Fraction:
92     """Return sum_y max_s pi(s) C(y/s) for a finite channel."""
93
94     if len(prior) != len(channel):
95         raise ValueError("One channel row is required per secret")
96     outputs = len(channel[0])
97     if any(len(row) != outputs for row in channel):
98         raise ValueError("Channel rows must have equal width")
99     return sum(
100         max(prior[s] * channel[s][y] for s in range(len(prior)))
101         for y in range(outputs)
102     )
103
104
105 def self_check() -> None:
106     examples = (
107         ((0, 0),),
108         ((1, 2), (0, 3)),
109         ((2, 1), (1, 0), (3, 2)),
110         ((0, 3), (1, 2), (2, 1), (3, 0)),
111     )
112     for counts in examples:
113         assert transfer_score(3, counts) == path_score(3, counts)
114
115     vectors = ((9, 117649, 1), (1058841, 1, 117649),
116               (3969, 1750329, 35721))
117     assert pareto_frontier(vectors) == tuple(sorted(vectors[1:]))
118
119
120 if __name__ == "__main__":
121     self_check()
122     print("exact hidden-state core: PASS")

```

## E Reproduction Commands and Artifact Manifest

With Python 3.11 or newer and numpy>=2.0, run the release checks from the extracted prismatic-fourier-lab-v0.10.0 directory:

```

python -m compileall -q ./src ./tests ./run_demo.py
python ./run_demo.py --milestone 10
python -m unittest -v tests.test_milestone10
python -m unittest discover -s ./tests -v
python ./run_demo.py --milestone 10 --json > milestone10-report.json
.\build.ps1

```

The exhaustive phase can remain silent for hours on some machines. Expected terminations are OVERALL STATUS: PASS, 12/12 dedicated tests, and 126/126 complete tests.